Earth and Planetary Science Letters, 82 (1987) 180–192
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Paleomagnetic evolution of the Central Massif (France) during the Carboniferous

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Received May 29, 1986; revised version accepted December 2, 1986

The present paper aims to synthesize results of a systematic paleomagnetic investigation performed on metamorphic, plutonic and volcanic series from the Central Massif. Detailed, thermal and alternating field demagnetizations yield a large set of paleomagnetic directions. Several groups of directions corresponding to different age intervals are identified. The group D mean direction: $D = 288°$, $I = 57°$ (37°S, 110°E), characterizes Late Devonian/Early Carboniferous metamorphic and plutonic rocks from Limousin. The group C′ directions: $D = 301°$, $I = 24°$ (30°S, 79°E), represent Late Visean/Namurian magnetizations, present in the major investigated areas. The group B directions: $D = 249°$, $I = 7°$ (12°N, 111°E), exist not only in the whole Central Massif, but also in other Paleozoic outcrops of the Variscan belt. They were acquired during the Namurian/Westphalian. The group A′-A directions are the only typically "European" magnetic directions. They have taken place in Stephanian/Autunian times, mainly during the Kiaman reversed interval. Interpretation of these directions in terms of geodynamics leads to a probable large S-N drift of the massif during the Latest Devonian/Early Carboniferous followed by two important rotation phases, first in the Middle Carboniferous, then at the end of the Westphalian. These rotations have also affected other massifs of the Variscan belt.

1. Introduction

In spite of numerous geological and paleomagnetic studies, the plate and microplate motions that have given rise to the Variscan belt are still ill understood. Tarling has recently exposed the problems and limits of Paleozoic paleomagnetism [1]. The mean Devonian and Carboniferous virtual paleopoles of northern Europe are very affected by Late Variscan overprints, and the commonly accepted paleopositions of Europe during Hercynian times are probably not definitive [2]. In the same way, the latest paleomagnetic models proposed for the Variscan belt cannot be considered as satisfying because they are often based on ill-dated remagnetizations [3,4]. Concerning the African plate, the lack of data is dramatic. Recent paleomagnetic and radiometric studies on the so-called "Msissi norite" in southern Morocco reject the unique Late Devonian paleopole of Africa [5] and question the existence of a widespread Hercynian ocean.

To constrain the evolution of the Hercynides, we have undertaken a systematic paleomagnetic investigation of the belt itself, from the Armorican Massif to the Sudetians. This study aims at discovering local and regional deformations and deciphering the global motion of the chain during the Late Devonian/Early Permian time-span. Around 300 Ma, important changes in declination, probably related to the Asturian phase, have been found in the Corso-Sardinian microblock [6]. Evidence for Carboniferous rotations exists also in the Vosges [7,8], in the Ardenne Massif [9] and in the Armorican Massif [10]. The present paper attempts to synthesize already published results on the Central Massif [11–13], together with recent unpublished data, in order to propose a model of evolution for the massif during Variscan times. The model is then compared with results from other Paleozoic massifs.

2. Sampling and laboratory techniques

Sampling has been performed on various volcanic, plutonic and metamorphic series of the western, northern and northeastern Central Massif (Fig. 1). Whenever possible the samples were col-

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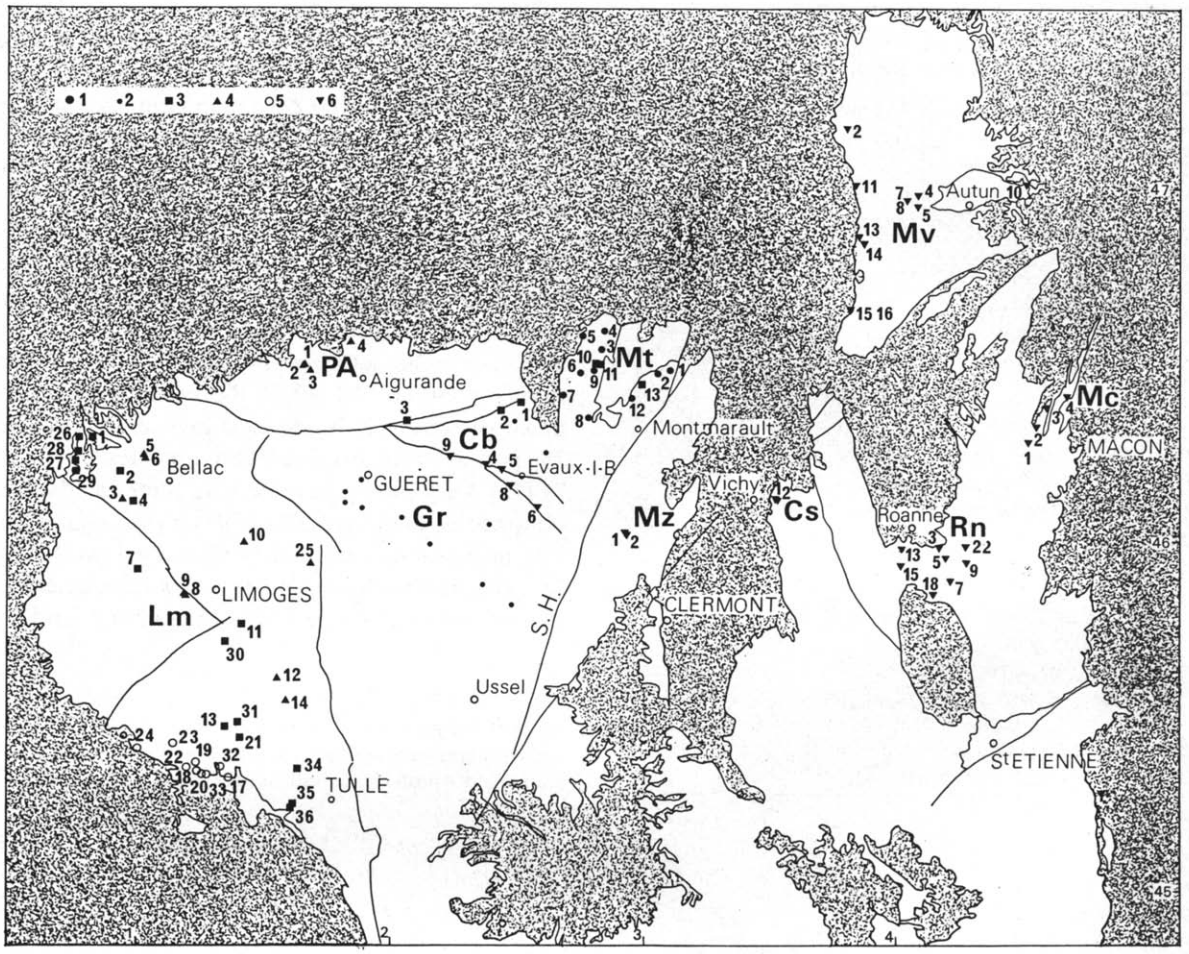


Fig. 1. Location of the paleomagnetic sites. *1* = granites; *2* = Gueret granite, sampled by Bouchez et al. [39]; *3* = diorites and tonalites; *4* = amphibolites; *5* = metamorphic series of Bas Limousin; *6* = Carboniferous volcanics; *Lm* = Limousin; *PA* = Aigurande Plateau; *Cb* = Combrailles; *Mt* = Montmarault; *Mz* = Manzat; *Cs* = Cussey; Rn = Roannais; *Mv* = Morvan; *Mc* = Maconnais.

lected from radiometrically dated sites. About 90 sites were investigated, but only 2/3 of them contain characteristic magnetizations. A site represents an exposure of at least 10 m^2. A quarry cutting a unique rock formation is taken as one site. Usually, 5–10 oriented samples were drilled or collected as blocks. From each sample at least 3 specimens of 2.5 cm diameter and 2.2 or 2.5 cm length were obtained. Remanent magnetization measurements were performed with a Digico magnetometer, with a noise level reduced to 3×10^{-5} A m^{-1}.

Paleozoic rocks are often characterized by multicomponent magnetizations. In order to isolate them, detailed stepwise thermal and alternating field demagnetizations were undertaken on all samples. In most cases, the thermal process was the most efficient. Tectonic corrections were only possible in Carboniferous, layered, volcanic series. The dip was measured on sedimentary hostrocks, on flames in ignimbrites and ignimbritic tuffs, and on the foliation determined in laboratory from anisotropy of susceptibility measurements.

3. The new results

The recent investigations complete a previous study on Paleozoic series of the Limousin [13]. The existence of magnetizations with high inclinations, probably acquired during the Late Devonian/Early Carboniferous had to be supported by additional data.

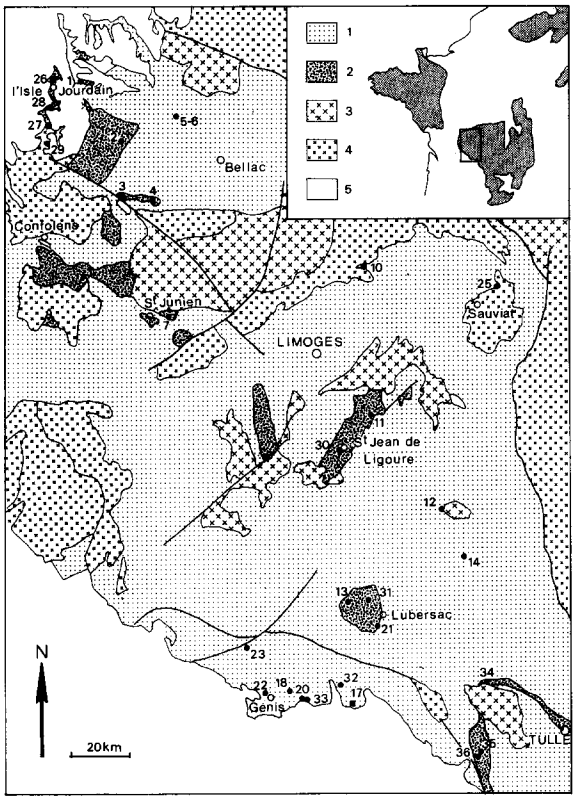


Fig. 2. Geological sketch map of the Limousin and location of the paleomagnetic sites. *1* = metamorphics; *2* = tonalites; *3* = Early Carboniferous granites; *4* = Middle/Late Carboniferous granites; *5* = post-Visean sedimentary cover.

The investigated series are tonalites (Lm 26, 27, 28, 29, 30, 31, 34, 35, 36), emplaced during the Late Devonian and Cambro-Silurian volcanics metamorphosed in the Devonian (Lm 32, 33) (Fig. 2). Subsequently to the emplacement of the tonalites, the Limousin area has not undergone important tectonic phases.

After thermal and alternating field demagnetizations, the tonalites generally display two types of stable magnetic components. Those with maximum unblocking temperatures up to 580°C (magnetite) are characterized by very scattered directions which do not allow the computation of a mean direction. It is the case in sites Lm 26, 28, 29, 30 and in the previous study, in sites Lm 2, 11, 21 [13]. Very stable characteristic components were obtained in the range 570–630°C, with high negative inclinations that will be discussed later (Fig. 3). The metamorphic rocks show characteristic magnetizations carried by magnetite and/or

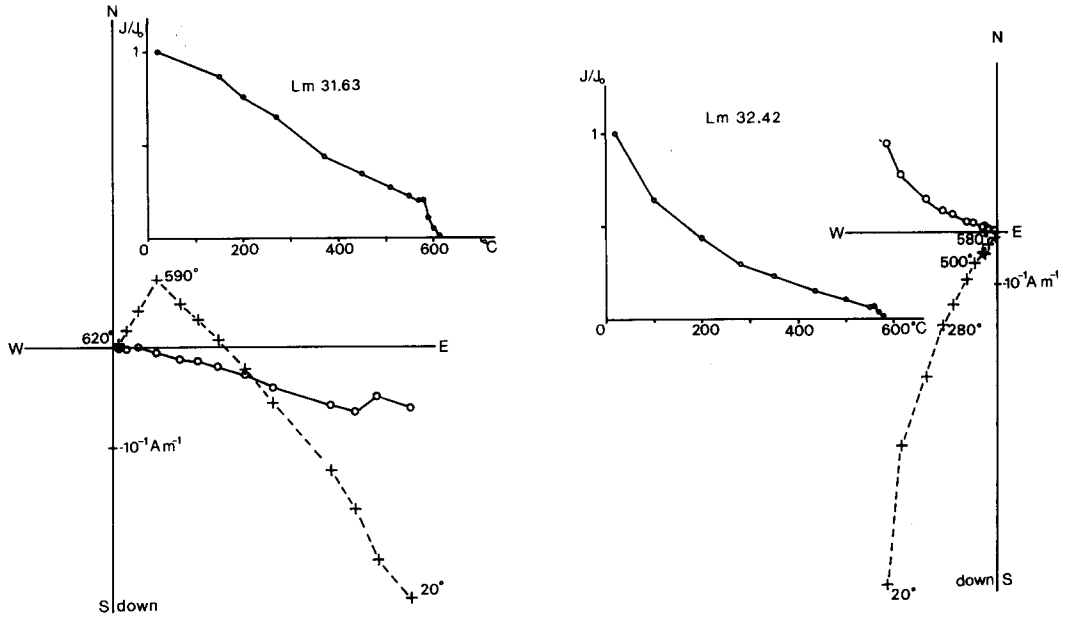


Fig. 3. Typical vector diagrams of thermal demagnetizations. Lm 31.63 is a tonalite; Lm 32.42 a metabasalt. Orthogonal projections in the horizontal plane (○) and in the E-W (+) vertical plane. J_0 is the intensity of NRM at room temperature, J, the intensity after heating.

TABLE 1

Characteristic directions of magnetizations in tonalites of Limousin and metavolcanics of the Genis area.

Site	Locality	Rock type	T_B (°C)	N/N_0	n	D (°)	I (°)	k	α_{95} (°)
Lm 1	Adriers	tonalite	500–580	5/7	11	326	78	22	10
			580	2/7	4	96	−65	148	8
*Lm 27	l'Isle Jourdain	tonalite	570–580	2/6	4	96	−52	503	4
Lm 3	Moulin du Pont	tonalite	580	2/6	4	198	8	50	13
		+ amphibolite	230–580	3/6	7	239	26	30	11
Lm 4	Nouic	tonalite	300–400	5/7	8	272	73	37	9
			500	5/7	9	302	−15	10	17
Lm 7	St Junien	tonalite	620	6/10	12	198	−3	46	6
			580	4/10	8	103	−19	64	7
			650	2/10	3	74	−21	306	7
Lm 13	St Julien-	tonalite	600–630	6/6	12	103	−66	80	5
*Lm 31	le Vendomois	tonalite	620	8/8	10	108	−50	59	6
*Lm 34	Joujoux	tonalite	620	8/9	13	306	−63	25	8
*Lm 35	Ste Féréole	tonalite	620	3/3	7	148	57	43	9
*Lm 36	Ste Féréole	tonalite	620	3/7	7	155	−76	67	7
			580	3/7	4	287	−6	88	10
			580	2/7	4	260	−1	34	16
Lm 18	Genis	ignimbrites	580	7/9	11	248	10	64	6
			580	4/9	5	266	38	42	12
Lm 20	Genis	ignimbrites	580 (1)	7/8	8	284	53	33	10
			580 (2)	6/8	7	230	49	55	8
			580 (1 + 2)	7/8	15	258	55	15	10
			580	3/8	4	217	6	56	12
*Lm 33	Genis	ignimbrites	580	7/7	12	29	59	115	4
			670	6/7	9	301	51	52	7
Lm 22	Genis	gabbro	580–590	3/6	5	311	43	112	7
			580–590	5/6	7	247	10	39	10
Lm 23	Genis	dolerite	350–550	4/7	6	237	19	22	14.5
*Lm 32	Genis	metabasalt	580–600	3/7	5	289	26	38	13
			580–600	5/7	10	284.5	61.5	42	8
Lm 17	Thiviers series		350–400	4/6	4	175.5	3	83	10

Asterisks denote new results; other data from Edel [13]. T_B = maximum unblocking temperatures; N = number of samples taken in the statistic; N_0 = total number of samples; n = number of specimen taken in the statistic; D, I = in-situ declination, inclination; k, α_{95} = Fisher statistic parameters.

hematite, and directions nearly opposite to the latter (Fig. 3). All characteristic directions are listed in Table 1, together with the results obtained previously in the same series.

4. The paleomagnetic directions

The stereograms of Figs. 4, 5, 6 represent all mean directions obtained for the different magnetic components. The scattered results imply a rather complicated paleomagnetic evolution of the Central Massif. Despite of this scatter, several clusters can be observed. They have been defined as follows, each group having a corresponding inverse grouping:

group D: WNW declinations, high positive inclinations;

group C′: WNW declinations, low positive inclinations;

group C: NW declinations, low negative inclinations;

group B: WSW declinations, low positive inclinations;

group A′: SSW declinations, low positive inclinations;

group A: SSW declinations, low negative inclinations.

Because of the low inclinations of the A, A′, B and C′ directions, an equatorial cylindrical projection

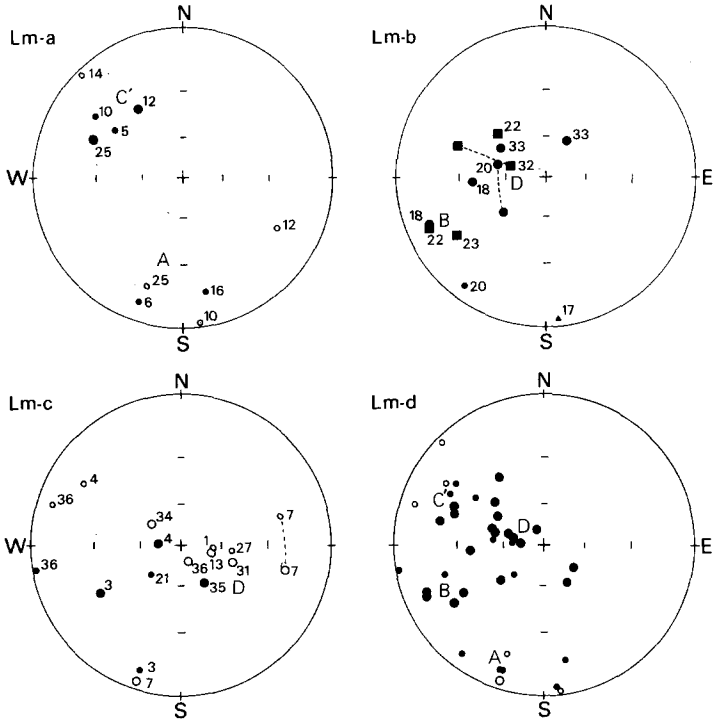


Fig. 4. Stereographic projection of the mean site paleomagnetic directions obtained in the Limousin. *Lm-a*: amphibolites, *Lm-b*: metaignimbrites (dots) and other metavolcanics (squares), *Lm-c*: tonalites. *Lm-d* shows all directions after appropriate polarity inversion and the main groups of directions. Small symbols represent directions computed from less than 3 samples, whatever the number of specimen is. Dashed lines significate that a trend exists between two directions.

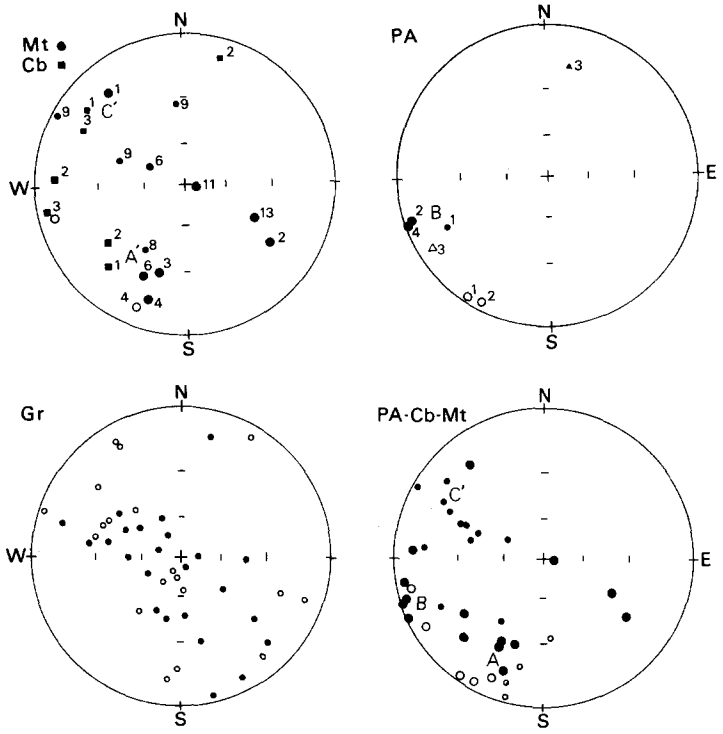


Fig. 5. Stereographic projection of the site mean directions obtained in the northern Central Massif. *Mt*: Montmarault granites and dioritic inclusions, *Cb*: Combrailles gabbro-diorites, *PA*: Aigurande Plateau amphibolites, *Gr*: shows the specimen directions measured in the Gueret granite. The samples were collected by Bouchez et al. [39] for anisotropy of susceptibility measurements. Because of the small number of specimens per site (< 3), each symbol represents one specimen direction. In stereogram *PA-Cb-Mt* are plotted all site mean directions, including *Cb* of Fig. 4, after appropriate polarity inversion.

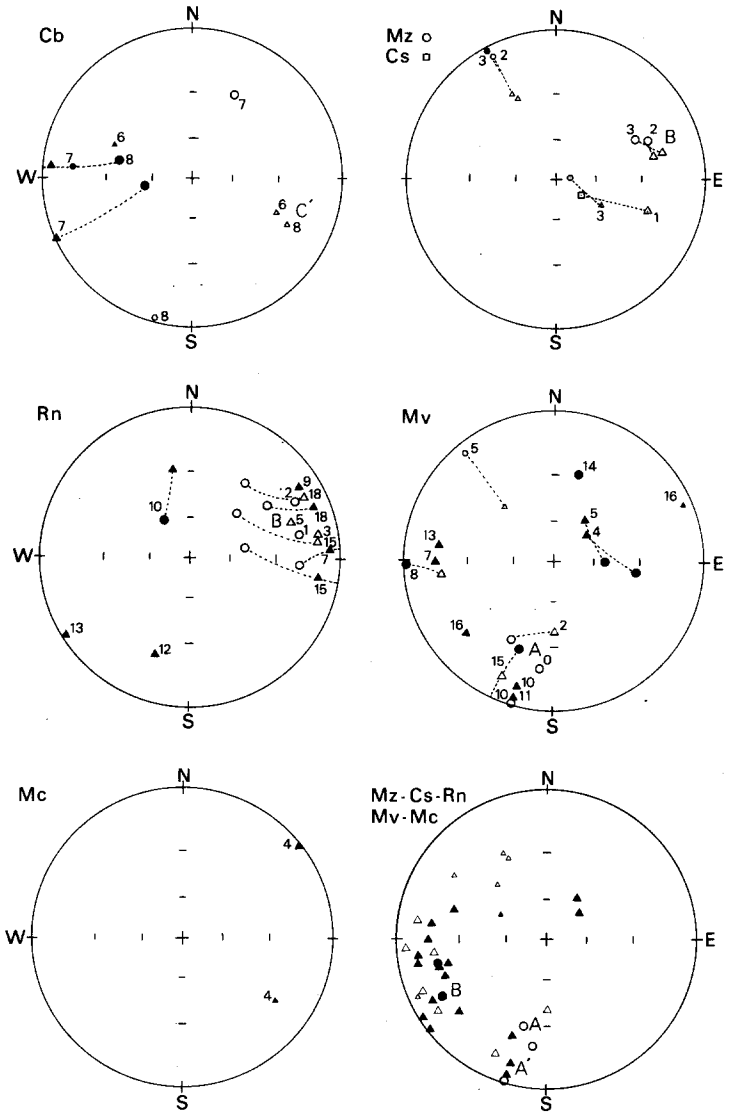


Fig. 6. Stereographic projection of the site mean directions measured in Carboniferous volcanics of *Cb*: Combrailles, *Mz*: Manzat, *Cs*: Cussey, *Rn*: Roannais, *Mv*: Morvan and *Mc*: Maconnais, before (dots) and/or after tectonic correction (triangles). The last stereogram shows all directions obtained in the Carboniferous volcanics after inversion of the easterly directions.

is more demonstrative of the different clusters than a polar stereographic projection (Fig. 7).

4.1. Group D directions

These are only found in pre-Carboniferous series of the Limousin area (Lm-b, c, d, Fig. 4). The Middle/Late Cambrian to Silurian meta-volcanites (LM 32, 18, 20, 33) and metagabbros (Lm 22) of Bas Limousin display a clear cluster around: $D = 289°$, $I = 50°$, $\alpha_{95} = 14°$ (Lm-b, Fig. 4). The blocking temperatures are high: 580 and/or 670°C. The metamorphism lasted from the Silurian to the Devonian/Carboniferous boundary (360–350 Ma) [14–16]. During the Carboniferous and later, the investigated area did not undergo significant deformations and the nearest Carboniferous granites are located 30 km away. Consequently, one can expect that the D magnetizations are at least 350 Ma old.

Directions nearly antiparallel to these positive D directions exist in the tonalites (or quartz-diorites) of the Limousin. They crop out along a

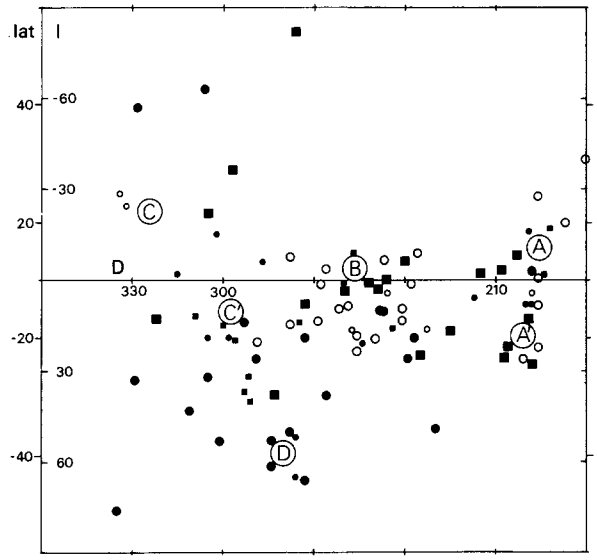


Fig. 7. Equatorial cylindrical projection of the mean paleomagnetic directions, illustrating the A, A′, B, C′, C and D groups, after inversion of the easterly directions, in Limousin (full circles), in northern (open circles) and northeastern Central Massif (squares).

N150° line extending from l'Isle-Jourdin (west of Bellac) to the Tulle region (Fig. 2; Lm-c, Fig. 4). Five sites (Lm 1, 13, 27, 31, 36) yield a mean direction: $D = 106°$, $I = -63°$, $\alpha_{95} = 13°$. During demagnetization they appear only in the range 570–630°C whereas the magnetizations with unblocking temperatures up to 580°C exhibit very scattered directions. K-Ar ages available for sites Lm 1 and Lm 13 are respectively 353 ± 10 Ma and 336 ± 9 Ma [11]. In fact, the tonalites were emplaced during the later Devonian as evidenced by the most reliable ages of 379–353 Ma [14,15,17]. Younger dates around 320 Ma are attributed to local Hercynian tectonometamorphic events by the same authors. The 620°C maximum unblocking temperatures that characterize the reversed D magnetizations suggest deuteric oxidation. Consequently, the magnetizing process may have been contemporaneous of the cooling of the tonalites, i.e. in the range 379–353 Ma.

In the tonalites as well as in the Bas Limousin metamorphics, the age of 350 Ma seems to be the upper limit for the D directions. The mean direction computed from D normal and reversed directions: $D = 288°$, $I = 57°$, $\alpha_{95} = 9°$ ($N = 10$), yields a virtual geomagnetic pole at 37°S, 110°E. Assuming that no important tectonic tilting has affected the Limousin during the Carboniferous, this result leads to a paleolatitude of the southern Limousin at 38°S (or 38°N) in the Latest Devonian.

4.2. Group C′-C directions

These are characterized by WNW declinations close to the D declinations but with lower inclinations. The C′ magnetizations were encountered in different Paleozoic series up to the Late Visean. In Limousin, they exist in amphiboles and some tonalites (Lm-a, c, d, Fig. 4). The amphibolites of site Lm 10, lying close to the 326 ± 18 Ma old granite of St. Sylvestre [18], exhibit a K-Ar age of 314 ± 9 Ma [14]. The Sauviat eclogitic complex (Lm 25) shows an age of 320^{+25}_{-17} Ma [19] which is probably related to the intrusion, around 332 Ma, of the neighbouring Millevache granite [18]. The St. Junien tonalite (Lm 7), also characterized by a C′ direction but also by B and A components produced similar ages of 313 Ma (Rb-Sr [5]) and 326 ± 10 Ma (K-Ar [14]). More to the north, in the plutonic massifs of Gueret (Gr, Fig. 1) and Montmaraults (Mt, Fig. 1), the scattered directions are mainly situated on a great circle oriented N300° (Fig. 5). As the granite of Gueret was emplaced at the Devonian/Carboniferous boundary (356 ± 10 Ma [20]), the scatter is probably due to badly demagnetized C′ + D (normal and/or reversed) multicomponents. Because the magnetic viscosity increasing above 300°C, the Gueret directions represent only means obtained in the range 20–300°C. The sub-alkaline Montmarault granite and its dioritic inclusions are not dated.

C′ directions were also found in Late Visean tuffs of Combrailles (Cb, Fig. 6) and in the Cussey area, east of the Limagne graben (Mz-Cs, Fig. 6). After tilt correction, the normal and reversed directions of site Cb 6, measured in two separate flows, become nearly antiparallel. This positive fold test indicates that the magnetizations were emplaced before folding, i.e. during the Late Visean. The tuffs of site Cs 1 have exhibited ages of 336 ± 5 Ma (Rb-Sr) and 327 Ma (K-Ar) [21]. They display a reversed D direction before tilt correction and a reversed C′ direction after. The corrected C′ direction lies very close to those obtained west of the Sillon Houiller in sites Cb 6 and Cb 8. Summing up, the C′ directions exist in the western, northern and eastern Central Massif.

The mean direction computed from the most reliable results $D = 301°$, $I = 24°$, $\alpha_{95} = 8°$, for $N = 11$, and the corresponding pole 30°S, 79°E. The available dates lead to propose a Late Visean/Namurian age of emplacement (335–315 Ma).

Some northwesterly directions show negative inclinations (and conversely) (Mt 2, 13: Mt-Cb, Fig. 5; Mz 2, 3; Mz-Cs, Fig. 6; Mv 5, Fig. 6). Such C directions are very common in Visean volcanics and plutonics of the Vosges Massif. The corresponding magnetizations were acquired during the 345–325 Ma interval [7,8] (Fig. 8). C directions appear also in the Spessart (F.R.G.), where they are probably related to the cooling of the latest metamorphic phase, i.e. between 335 and 325 Ma [22,23]. In the present stage of knowledge, it is unclear if the rare C directions observed in the Central Massif are deviated C′ directions or if they represent a poorly recorded stage of the paleomagnetic evolution of the massif.

4.3. Group B directions

The WSW and ENE B directions are present in all investigated areas and in very different rock types, metamorphics, plutonics and volcanics (Figs. 4, 5, 6, 7). In southern Limousin the mean B direction: $D = 248°$, $I = 15°$, $\alpha_{95} = 10°$ (10°N, 113°E), is obtained from Lower Paleozoic metavolcanics and tonalites (Lm 3, 7, 18, 22, 23, 36). Except for site Lm 7 that exhibits also C′ and A directions and was dated at 326 Ma (K-Ar) and 313 Ma (Rb-Sr), there is no indication as to the age of emplacement of these components.

Typical B directions characterize the amphibolites of the Aigurande Plateau in the north of the Central Massif (PA, Fig. 5). Metamorphism occurred during the Devonian. K-Ar ages range from 307 to 470 Ma, with a cluster around 315 Ma (306, 315, 320, 322 [4]). These younger ages coincide with the date of intrusion of leucogranitic laccoliths into the metamorphic basement (312 ± 6 Ma [18]). As three of the sites were collected close to such a granite, it is tempting to correlate the creation of B magnetizations with the cooling of the laccolith. Otherwise, the rather low unblocking temperatures around 400°C indicate that the overprints happened at the end of metamorphism.

The B directions are particularly well represented in Carboniferous volcanites. West of the Limagne graben, they are present in the fine grained Late Visean Manzat tuffs (Mz-Cs, Fig. 6). In the east, they characterize most of the Middle/Late Visean tuffs of the Roannais (Rn, Fig. 6). Isolated B directions exist also in Combrailles (Cb 7), Morvan (Mv 16) and Maconnais (Mc 4, Fig. 6). When available the tectonic correction improves the cluster, but in several cases a partial correction also reduces the scatter. As the series were also affected by Tertiary graben tectonics, one can not exclude that a part of these magnetizations may have taken place during or even after the end of the post-Visean folding phase.

In two sites (Rn 15, 18), the magnetizations carried by hematite show directions deviated counterclockwise relative to those defined by magnetic blocking temperatures. As in these rocks hematite probably results from oxidation of magnetite, the change in declination indicates a clockwise rotation of the series relative to north between both magnetization phases. The mean B direction computed for the Roannais volcanics is $D = 251°$, $I = 4°$, $\alpha_{95} = 12°$ for $N = 8$ (12°N, 109°E) [11,24]. In the Morvan, the Devonian volcanics of site Mv 16 have been affected by the emplacement of neighbouring granites, as exemplified by the 305 ± 9 Ma K-Ar date (Fig. 6) [25]. This age may correspond to both opposite B directions obtained in the freshest rocks. The less fresh samples (Mv 15) display secondary A directions.

In short, the B magnetizations were acquired during the Late Visean/Westphalian time-span. The mean direction for the Central Massif is: $D = 249°$, $I = 7°$, $\alpha_{95} = 6°$ ($N = 23$), and the mean VGP: 12°N, 111°E. While the Late Visean volcanics of the northeastern massif mainly show ENE directions, the metamorphic series from the western and northern part display only WSW opposite directions. As the latter are often characterized by lower unblocking temperatures, the WSW paleomagnetic period may be the younger.

Very similar results were observed in other Paleozoic outcrops in the Variscan belt. In the Armorican Massif, Tournaisian volcanics from the northern limb of the Laval Basin display post-folding i.e. post-Namurian, B directions [10]. Southwest of Great Britain, in Pembrokeshire, Old Red Sandstones exhibit Carboniferous/Permian remagnetizations with WSW declinations and low inclinations [26]. In the Ardenne Massif, B direc-

tions were obtained in volcanic and sedimentary series ranging from Cambrian to Visean. They are interpreted as remagnetizations emplaced during the Namurian/Westphalian folding phase, likely around 316 Ma [9]. We have also found B directions in Late Variscan rhyolites of the northern Vosges (B) and in ignimbrites of the Munstertal (Black Forest) dated at 307 Ma [2,27] (Fig. 8). These dates lead to restrict the B paleomagnetic event to the Namurian/Westphalian and to reduce the overlap with the C′ event.

4.4 Group A′-A directions

They correspond much more to conventional "European" directions than the previous components. The A′ directions show declinations around N195° and slightly positive inclinations. They exist in all of the investigated series, mostly as

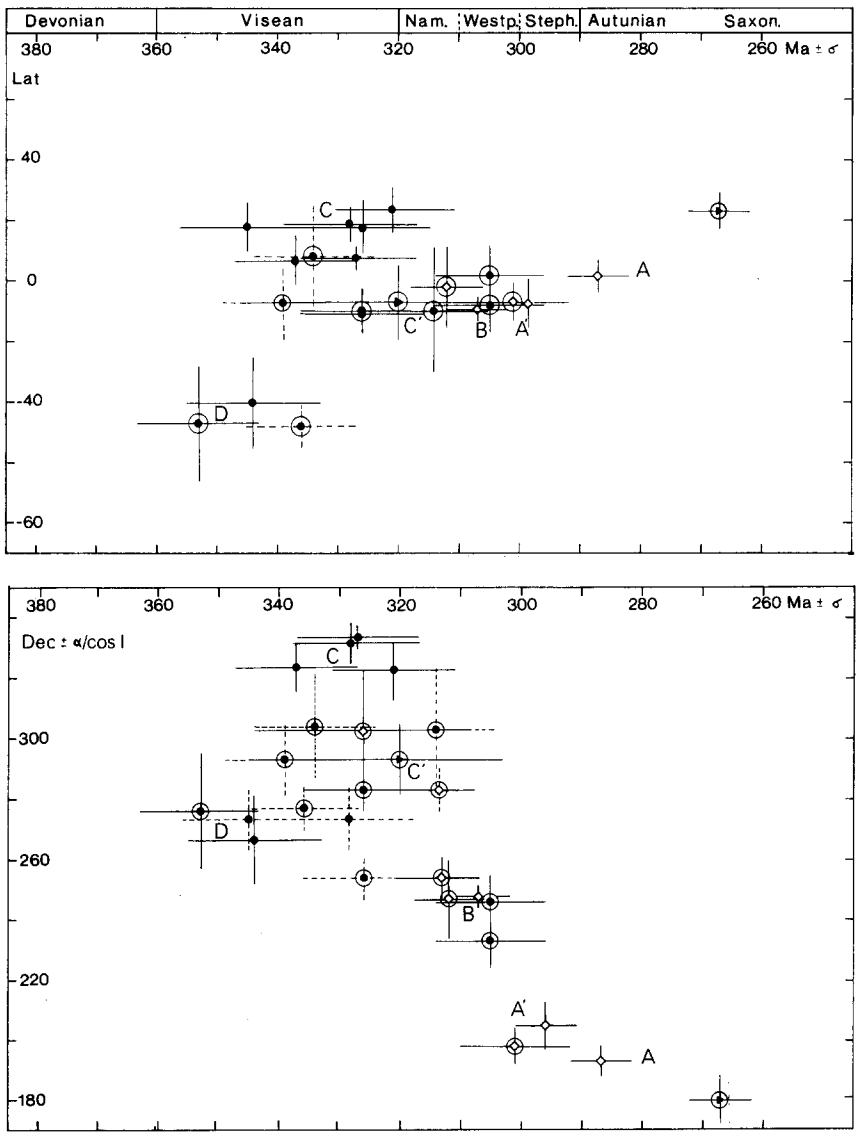


Fig. 8. Evolution of the declinations and paleolatitudes with the radiometric ages measured in Paleozoic rock formations of the Central Massif (circled symbols), the Vosges and the Black Forest, by K-Ar (dots), Rb-Sr (diamonds) and U-Pb (triangles) methods. Because of the uncertainty relative to the C′, C and D polarity, this representation is not unique. Here the declinations indicate the south. When two ages or two directions are available, the less probable solution is represented with dashed error bars.

TABLE 2

Group	N	D (°)	I (°)	α_{95} (°)	Lat.	Long.	Age
D	10	288	57	9	37°S	110°E	Late Devonian/Early Carboniferous (379–350 Ma)
C′	12	301	24	8	30°S	79°E	Late Visean (Namurian) (335–320 Ma)
B	26	249	7	6	12°N	111°E	Namurian/Westphalian (320–310 Ma)
A′-A	21	198	0	8	32°N	162°E	Stephanian/Autunian (300–260 Ma)

secondary magnetizations. In the Montmarault Massif, A′ is probably a primary direction $D = 199°$, $I = 13°$ in a leucogranite dated at 301 ± 9 Ma [28]. Very similar results appear in dioritic inclusions of the Montmarault granite collected in the vicinity of leucogranites of the same age (Mt 3, 6). The mean direction computed from the three sites: $D = 201°$, $I = 21°$, $\alpha_{95} = 13°$, is interpreted as representing the magnetic field around 300 Ma. Similar directions have been measured in Stephanian sandstones from the eastern Central Massif [29]. In Morvan, the A directions were recorded during emplacement of Autunian ignimbrites dated at 267 ± 5 Ma [30] and in lamprophyres intrusive into the ignimbrites (Mv 2, 10) [11,24]. They are present as overprints in all earlier series.

Out of the Central Massif, the A directions characterize Stephanian/Permian sandstones and Early Permian volcanics (mean VGP: 43°N, 168°E [31]). In the Black Forest we have identified A′ primary directions in porphyries dated at 296 Ma, and A directions in slightly younger extrusions emplaced at 287 Ma [2,27] (Fig. 8).

5. Paleomagnetic evolution of the massif

One can infer that the different groups of directions correspond to different age intervals. The overlapping of the C′ and B periods is due to the poor dating of the Late Visean volcanics and of the Sudetian II tectonic phase (Latest Visean/Namurian). Comparisons with neighbouring massifs lead to the following sequence (Fig. 8) (Table 2). The existence of B to A directions with consistent ages demonstrates that, from the Namurian to the Permian, the Central Massif had a homogeneous paleomagnetic history. C′ directions are present in the western part of the massif and, west of the Sillon Houiller, near Cussey. Some local rotations are not excluded. For instance, the N270° directions observed in the Morvan (Mv 7, 8, 13, Fig. 6) could be counterclockwise rotated C′ directions or clockwise rotated B directions. In the first case, the counterclockwise rotation of the series could result from the Sudetian phase. Similar considerations can be applied for the gabbro-diorites of the Boussac series, located between two E-W strike faults (Mt 2, 3, Fig. 5).

Thanks to the presence of D, C′, B, A′ and A directions in Limousin, an attempt can be made to model the geodynamic evolution for this region. In view of the uncertainty relative to the polarity of the different paleomagnetic directions, several different models can be developed. With exception of the SSW A and A′ directions which indicate the paleosouth direction, there is an undetermination for the other directions. Several arguments are in favour of a reversed polarity of the WSW B directions. First, a 50° clockwise rotation needs less energy than a 130° counterclockwise rotation, and this in a rather short period of about 10 Ma (310–300 Ma). In Roannais, a clockwise rotation of the series was observed during the interval defined by the emplacement of two B magnetizations, the first carried by magnetite, the second by hematite. As geological considerations exclude a local rotation, it is tempting to attribute this deviation to a more general motion, corresponding partially to the 50° clockwise rotation. In other massifs of the Variscan belt arguments exist in favour of Late Hercynian clockwise rotations, e.g. in the Armorican Massif, the Ardenne, the Vosges and the Black Forest [2,8–10].

With the assumption that the WSW B directions indicate the magnetic south direction in Namurian/Westphalian times, four solutions for the south direction wandering path remain to be discussed (Fig. 9). Solution (b) implies a very large N-S drift of about 140° of the Limousin during the Latest Devonian/Early Carboniferous (D^- $-C'^+$), with unrealistic spreading rates of 25–100 cm/yr. Solution (c) needs a 120° rotation from C'^- to B during the Late Visean/Namurian, corresponding to an important counterclockwise rota-

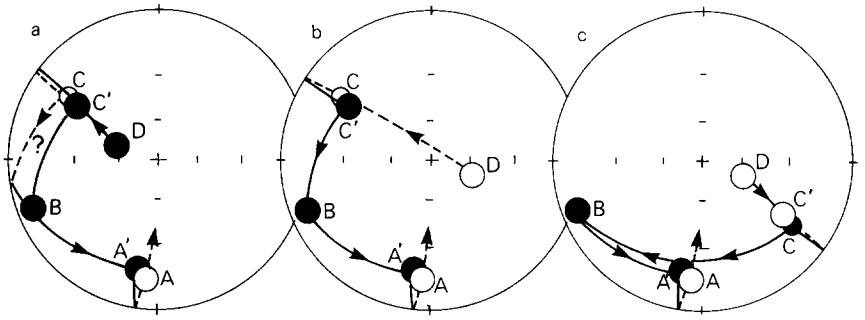


Fig. 9. Three possible south direction wander curves of Limousin during the Late Devonian/Early Permian time span. Stereographic projection. Open circles represent negative inclinations, full circles positive inclinations.

tion of the massif. The fourth solution is a combination of (b) and (c) and is excluded. Remains the less stringent solution (a), solution (c) being the next possible case. Subsequent to the Late Devonian, the Limousin would have undergone a significant northward drift of either 4000 km (D^+–C'^+) or 6000 km (D^+–C'^-. Assuming respective ages of 360 Ma and 325 Ma for the D and C′–C directions the spreading rates of 11 and 17 cm/yr are of the same order as that of India during its Cenozoic northward drift.

From a geological point of view, it is commonly admitted that in the Carboniferous the Hercynian ocean was closed [32,33]. The difference by about 15° between the paleolatitude of 38°S corresponding to the Limousin D directions and the mean paleolatitude computed from the Devonian/Early Carboniferous poles from northern Europe [1] is in contradiction with the geological observations. But one has to keep in mind that the mean Devonian/Carboniferous poles are highly influenced by Late Variscan overprints. A plot of the Devonian poles of Europe shows a main cluster around the Late Carboniferous/Permian poles [1]. If one takes into account the only poles situated in the new Guinea area, as suggested by Tarling [1], then the paleolatitudes of the Limousin and of northern Europe fit together. In this case, the large northward drift during the Early Carboniferous may have affected both blocks. Concerning the relations with the African plate, the lack of paleomagnetic data yields no constraints.

After acquisition of the C′ directions, the Central Massif has begun to rotate clockwise. An initial 50° rotation brought the massif into a quasi N-S trending position, corresponding to the B direction. A second rotation, also by about 50°, occurred in the 310–300 Ma time span. During the Late Carboniferous the Central Massif remained slightly south of the equator, crossing it probably around 290 Ma.

Despite the uncertainty relative to the polarities of the different paleomagnetic directions, the interpretation in terms of geodynamics leads to the following motions: (1) a large S-N or N-S drift; (2) a first rotation in the Middle Carboniferous; (3) a second rotation in the Late Carboniferous.

6. Correlations with other massifs of the Hercynides

The existence of B directions in the Armorican Massif, in Pembrokeshire, the Ardenne Massif, the Vosges and the Black Forest demonstrates that, in Namurian/Westphalian times, the Variscan belt was trending N-S. Consequently the whole belt has undergone the Late Carboniferous rotation. This has some consequences for the present knowledge of the paleopositions of "stable Europe" [2].

The Vosges and the Spessart display C directions that reveal an earlier rotation of these massifs in the Late Visean/Namurian [8]. Fig. 8 shows the similarity of the paleomagnetic behaviour of the Vosges, the Black Forest and the Central Massif. With possible slight relative motions, they all had the same geodynamic evolution after the Late Visean.

Directions that may correspond to those of group D exist also in the Middle/Late Devonian Beja Gabbro in southern Portugal [32]. After a 35° declination correction, replacing Iberia in its pre-Jurassic position, some of the scattered Beja directions fall into the group D of the Limousin. One may infer that in Late Devonian times the

Limousin and southern Portugal were already in close relative positions as nowadays.

Directions similar to the D and C′ northwesterly directions were found in Precambrian to Cambrian plutonics and volcanics of the Armorican Massif and of Jersey (St. Quay diorite, Keralain gabbro [35]; Treguier kerotophyre [36]; Jersey dolerites [37]) and as high-temperature magnetizations in Cambro-Ordovician series of the Armorican Massif [38]. In Tournaisian volcanics of the Laval Basin, east of the massif, a few high-temperature magnetizations were measured with C′ directions before and C directions after tectonic correction [10]. This result, together with the similar directions in later Devonian/Early Carboniferous formations of the Central Massif and in Precambrian to Ordovician series of the Armorican Massif, question the age of the latter magnetizations. The Variscan orogeny has also affected the northern part of the massif and may have been responsible for overprint effects. It this is the case then the Armorican Massif may have had a similar evolution as the Central Massif.

Acknowledgements

Thanks are due to the geologists who have collaborated to this work by their fruitful advice and guidance in the field: M. Coulon, J.P. Floc'h, J. Grolier, J.M. Quenardel, P. Rollin and M. Turland. M. Lacaze began the study in Roannais and Morvan. Drs. M. Westphal and R. Montigny read the manuscript and made constructive comments. The work was supported by the ATP "Géodynamique" of the CNRS, the programme "Géologie Profonde de la France", and the laboratory UA 323 of the CNRS in Strasbourg.

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